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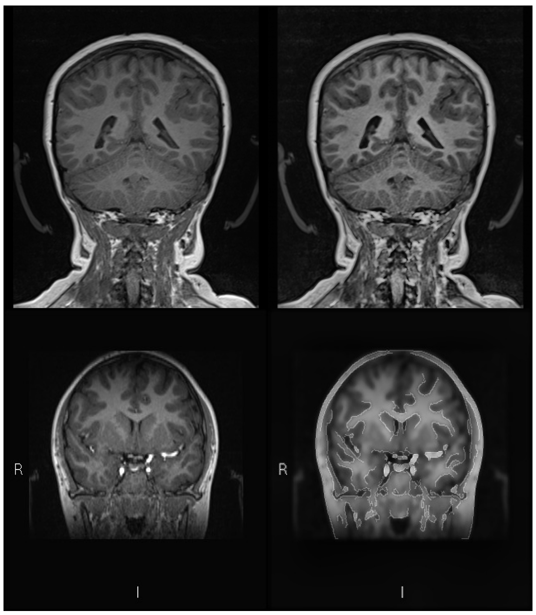
INSTITUT FOR MATEMATISKE FAG



Specialcouruse NRU

Holger Max Fløe Lyng

Investigating Methods for Polymicrogyria Classification

A replication of Guha et al. “Automated detection of polymicrogyria in pediatric patients using deep learning”

Dato: Juni 2026

Vejleder: Melania Ganz LLucia Coll Benejam

Dette er en skabelon for selve opstillingen. Kravene til hvilke oplysninger, der skal medtages, varierer fra fag til fag. Du kan derfor ikke uden videre benytte skabelonen til din egen afhandling. Du får nærmere oplysninger om kravene på dit eget studiekontor!

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Indholdsfortegnelse

1 [Purpose]

Article 2 - Preprocessing Reproduction (in progress). Reproduction of the preprocessing pipeline described in Article 2 is partially complete. Notably, the paper does not specify how differences in Field of View (FOV) or pixel/voxel spacing are handled across scans. As a first step, we will replicate their network architectures (ResNets and others) to verify whether we can reproduce the overfitting behavior reported in the paper.

2 Course motivation

2.1 Data problem

Data are split into HC and PMG cases. FOV is different within and between groups of HC and PMG cases. What excatly :::: check FOV, DPI, and JPG size
Cas

2.1.1 Article 1 zheng et al.

Table 1: Imaging parameters for the coronal T1 weighted sequence.

	Siemens 3T Skyra	General Electric 1.5T Cigna
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320×260	512×512
Slice thickness	1.2 mm	1.2 mm
FOV	20×23 cm	22×27 cm

Patients under the age of 18 with a diagnosis of PMG were identified from Childrens hospital of Eastern Ontario. A pediatric neuro radiologist with 12 years of experience in reading pediatric brain MRIs. If PMG presence was confirmed, the patient was included in the study. Images of the coronal 3D gradient echo T1 weighted sequence., high resolution sequence and high cortical details. Exported from PACS(Picture archiving and communication system) as JPEG. The cohort consists of 50% females with mean age of 12.3 years at the time of scan, STD 3.6 years, from 5.5 years to 17.8 years.

Dataset consists of 23 patients in total. within each patients MRI, nor all images show evidence of PMG and some slices are normal. Although the ratio between controls and patients is 3:1, the ratio between normal slices and anormals slices is 5:1. Each patients brain includes around 150 scans (full volumes?) on average.

2.1.2 article 2 guha et al.

Data description: Pediatric polymicrogyria MRI (PPMR) dataset by L. Zhang et aladd reff. The dataset consists of 23 patient MRI's, with a combination of PMG cases and healthy control cases.

Ratio between controls and PMG cases is 3:1, ratio between normal cases and anormaly cases is 5:1.

around 150 vrain scans of each patient were included in the dataset?? - Is this true 150 whole scans per subject?

patient cohort has 50 % of the female population?? Mean age of 12.3 during the scan process.

The normal group consists of patients who have shown symptoms of headaches and have undergone MRI scans.

The MRIs where provided in jpeg format from a coronal 3D gradient echo T1 weighted sequence. This dataset contains 15.056 total MRI images (jpeg(slices by slices)), 10.539 are control MRI and 4517 are PMG MRIs.

HMMM???

HC: add details PMG cases: add details

3 [Introduction]

Deep learning has transformed the analysis of medical images, achieving performance comparable to clinical experts across different tasks including diabetic retinopathy grading, skin lesion classification, and radiological triage [**?**]tilføj litjens ref. In neuroimaging, convolutional neural networks have been applied to the automated detection of structural brain abnormalities from MRI, with the promise of supporting radiologists in high-volume screening and reducing diagnostic delays in remote areas. However, this requires that reported performances metrics are grounded in sound methodology, free from data leakage, label noise, and low-level image confounds that can inflate accuracy without reflecting genuine pathology signal detection.

Polymicrogyria(PMG) is a malformation of cortical development characterised by excessive cortical folding and abnormal lamination[6]. It is one of the most common malformations of cortical development, accounting for approximately 20% of all such malformations [7]. PMG is frequently encountered in patients with drug-resistant and is commonly associated with neurodevelopmental delay, motor impairments, and cognitive deficits. PMG remains challenging to characterize reliably due to its heterogeneous radiological appearance, variable distribution of anatomical features, and the limited availability of curated neuroimaging datasets.

Deep learning attempts to classify PMG have predominantly been conducted on small, imbalanced cohorts and without proper patient-level data splitting[4], making it difficult to distinguish genuine classification performance from artifacts of data leakage, class imbalance, or low-level image difference introduced during acquisition. Furthermore, the use of models pretrained on natural image datasets — comprising everyday photographs of cars, birds, and household objects — introduces a substantial domain gap relative to clinical neuroimaging; ImageNet-pretrained features have been shown to transfer poorly beyond the lowest convolutional layers, offering little benefit over task specific training [**?**]tilføj raghu ref om domæne gap, which may further limit the sensitivity of these models to the subtle cortical morphology features relevant to PMG classification.

Automated detection of PMG from MRI could nonetheless inform clinical decision making by reducing the diagnostic burden on specialist neuroradiologists and enabling earlier identification of the condition. Fortunately, a publicly available benchmark dataset — the Pediatric Polymicrogyria MRI (PPMR) dataset [9] — has been released, and two independent studies have applied deep learning methods to PMG classification using this resource [9, 4].

This study undertakes an independent replication and methodological evaluation of guha et al. We examine the assumptions underlying their experimental design, assess whether the reported near perfect classification accuracy can be attributed to genuine PMG detection, and quantify the effect of applying corrective measure to the identified methodological issues. We identify three methodological errors in Guha et al.: (1) Label noise from including PMG-negative slices in the positive class, (2) a systematic class-correlated resolution difference that provides a shortcut signal, and (3) pre-split HC downsampling that produces artificially balanced evaluation sets and ignores patient-level integrity.

In addition, we evaluate the corrected pipeline on an independent clinical dataset of epilipsy patients, providing an assessment of generalization beyond the PPMR benchmark.

4 Background

- PMG background bases on `https://www.ncbi.nlm.nih.gov/books/NBK1329/andhttps://medlineplus.gov/genetics/condition/polymicrogyria/#resources`
- Maybe talk alittle about MRI
- Neural networks used. Resnet101 adn DenseNet201
- Maybe preprocessing steps or should that be en methods?

4.1 Polymicrogyria

Polymicrogyria (PMG) is a malformation of the developing brain characterized by abnormal cortical lamination and an excessive number of abnormally small gyri, resulting in an irregular folding pattern across all or part of the cerebral cortex [7][1]. PMG arises during the post-migrational stage of cortical development, distinguishing it from other malformations of cortical development (MCDs) such as lissencephaly, which affects the migration stage[6]. Multiple forms of PMG have been identified. Unilateral PMG affects one hemisphere only, whereas bilateral PMG involves both hemispheres, and severity ranges from mildly focal to extensive bilateral involvment. PMG accounts or approximately 20% of all MCDs, making it one of the most common brain malformations[7].

Despite its prevalence among MCDs, PMG remains challenging to diagnose reliably. MRI is the only imaging modality that provides sufficient soft-tissue contrast and spatial resolution to identify the fine cortical folds characteristic of PMG; computed tomography (CT) is inadequate for this task[6]. The radiological appearance of PMG is heterogeneous, e.g. features include an irregular or bumpy cortical surface, a stippled gray-white matter junction, and apparent cortical thickening. Some cases of PMG, adjacent microgyri fuse their overlying molecular layers, producing a deceptively smooth outer surface that can be mistaken for other MCDs. MRI appearance also changes with myelination, making diagnosis difficult in infants during the first year of life, whne low gray-white matter contrast may obscure the slight cortical irregularities of the condition.....

4.2 MRI

Magnetic Resonance Imaging (MRI) is the principal structural neuroimaging modality across a wide range of neurological conditions, including epilepsy, brain tumors, white matter diseases, and cortical malformations. Its clinical dominance in neurology stems from its capacity to provide high soft-tissue contrast without ionising radiation. T1-weigghted sequences in particular offer excellent differentiation between gray and white matter, enabling delineation of cortical thickness, sulcal morphology, and the gray-white matter junction architecture that is central to identifying MCDs such as PMG.

4.3 Deep learning for Medical Image Classification

4.3.1 Convolution Neural Networks

Convolution neural networks (CNNs) learn

4.3.2 ResNet-101

Residual learning was introduced by He et al. (2016)[**?**] to address the degradation problem in very deep networks, where accuracy saturates or degrades as depth increases. Each residual block learns a residual mapping $F(x)$ add more... so the block out is $H(x) = F(x) + x$. The skip connection allows gradients to flow directly through the network during backpropagation, making training of networks stable with over 100 layers. ResNet-101, comprising 101 weight layers organised in four residual stages achieved state-of-the-act performance on ImageNet ILSVRC 2015 and has become a widely adopted backbone for medical image and other classification tasks.

4.3.3 DenseNet-201

4.3.4 Transfer learning

4.4 Guha et al.

5 Methods

5.1 Data

5.1.1 PPMR

This project utilities the publicly available PPMR dataset introduced by [9] and subsequently used by [4], The dataset consists of coronal 3D gradient-echo T1-weighted MRI slices from 23 pediatric epilepsy patients with confirmed PMG and three age and gender-matched healthy controls per patient, yielding a 3:1 control-to-case patient ratio.

A critical property of the dataset is that not all MRI slices from PMG patients contrain visible malformation. Each slice was individually labeled by a pediatric neuroradiologist as PMG-positive (label 1), PMG-negative (label 2), or uncertain (label 3) [9]. As shown in Figure 1, only approximately 49% of all PMG-patient slices are labelled as PMG-positive; approximately 30% are PMG-negative, and approximately 19% are labeled uncertain.

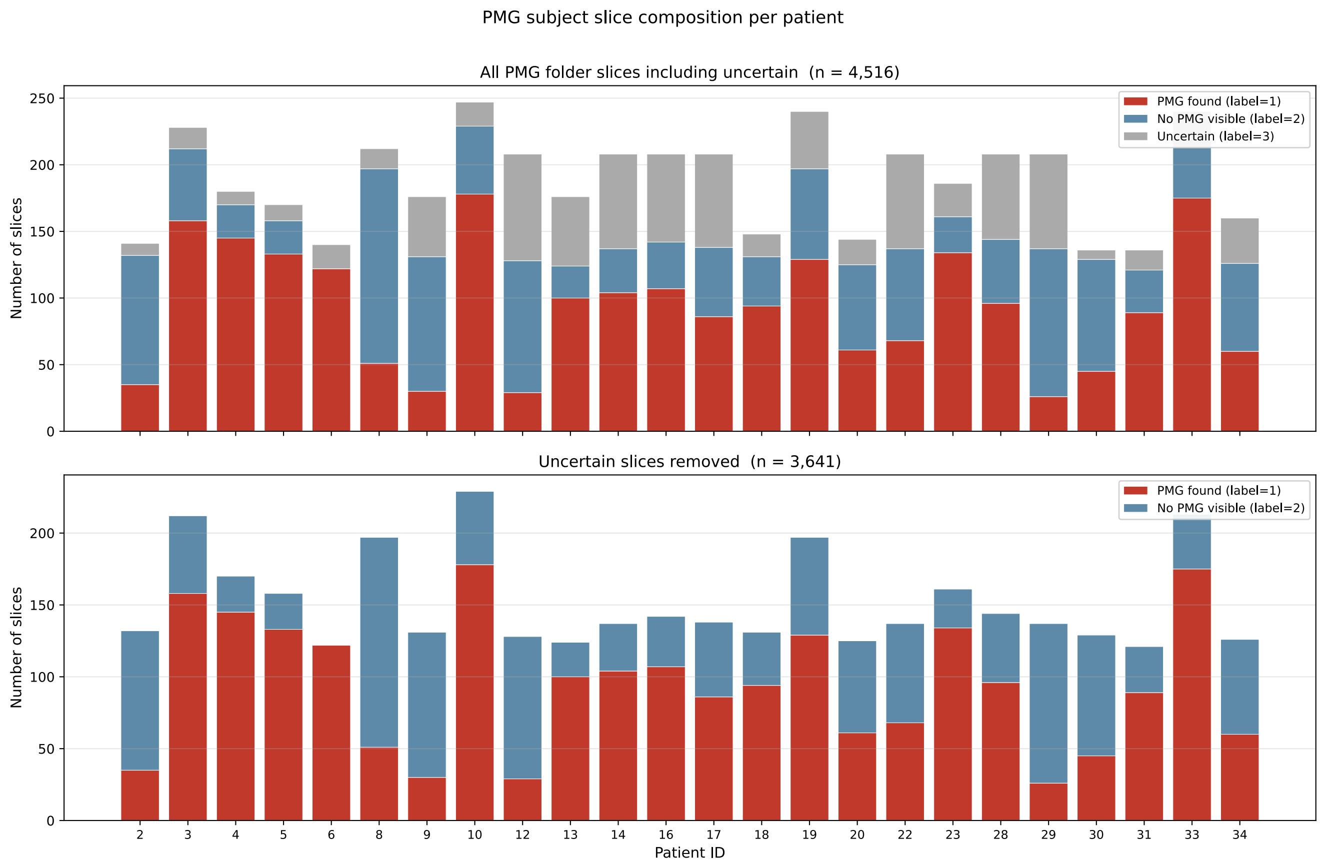

Figure 1: Top: Number of 2D slices per PMG patient labelled as PMG-positive (red), PMG-negative (blue), and uncertain (grey), based on the Kaggle release of the PPMR dataset [9]. Bottom: The same distribution excluding uncertain slices, replicating the figure published in the original dataset paper [9].

Aggregating across all labels, the dataset contains 4,517 attributed to PMG patients and 10,539 attributed to healthy controls. [4] report using precisely these counts in their analysis, treating all 4,517 PMG-patient slices as a single positive class and all 10,539 control slices as a single negative class. Here is the first methodological error: the positive class contains a large proportion of non-pathology slices, which can dilute the PMG-specific attributes and introduce label noise into both training and evaluation. In figure 2 are the full statistics for the dataset.

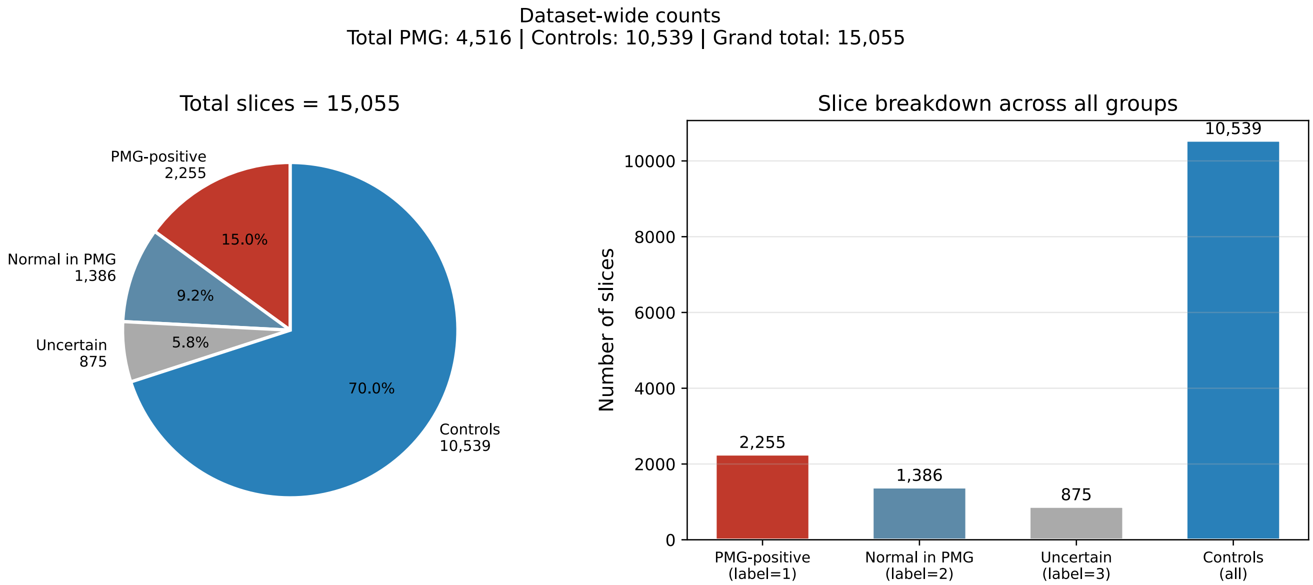

Figure 2: The full statistics of the dataset published on Kaggle[9]. Guha et al. used this data distr

The dataset was labeled by a pediatric neuroradiologist wth 12 of experience reading childrens brain MRIs. The patients was included in the study if PMG was found. The MRIs where a acquired using the highest resolution sequence for displaying cortical details, the coronal 3D gradient echo T1 weighted sequence, these volumes where exported as JPEG images from the PACS system. The imaging parameters used for acquisition of the coronal T1 weighted images are shown in Table 2

Parameter	**3 T Siemens Skyra**	**1.5 T GE Cigna**
Sequence	Coronal 3D GRE T1	Coronal 3D GRE T1
Repetition time (TR)	2200 ms	10.44 ms
Inversion time (TI)	1030 ms	450 ms
Echo time (TE)	2.63 ms	4.3 ms
Matrix	320×260	512×512
Slice thickness	1.2 mm	1.2 mm
Field of view (FOV)	20×23 cm	22×27 cm

Table 2: Acquisition parameters — PPMR dataset [9].

5.1.2 NRU data

To evaluate the preprocessing method and DL models used for the PMG classification task we introduce an in house dataset. The dataset includes 183 T1-weighted MRI scans from 162 unique patients evaluated for epilepsy (77 females; mean $\pm$ standard deviation (SD) age at acquisition: 25.8 ± 17.4 years). Of these, 110 scans (49 females; 24.7 ± 15.9 years) were reported to contain radiological evidence of PMG based on clinical radiology reports. The non-PMG group (37 females; 27.5 ± 19.6 years) consisted of scans in which PMG-related terminology appeared in the radiology report, but a subsequent double-rater confirmed the absence of PMG. Of these, 56 cases had other abnormalities and 17 were reported as having a normal MRI. The overall population distribution is shown in Fig. 1(c)which fig? ask melanie. MRI scans were acquired in routine clinical practice and retrospectively collected from the Eastern Region of Denmark, encompassing multiple scanner protocols, two magnetic field strengths (B), and three vendors. While no manual quality ratings were available, signal-to-noise ratio (SNR) was computed in white and gray matter as an objective measure of image quality. The mean $\pm$ SD SNR was 11.9 ± 2.7 in white matter and 7.9 ± 1.8 in gray matter, indicating that a variability in scan quality could contribute to errors in cortical surface reconstruction and feature extraction.

Parameter	**Value**
Scanner	Siemens Verio, 3 T
Sequence	MPRAGE (T1-weighted)
Repetition time (TR)	1900 ms
Echo time (TE)	2.23 ms
Inversion time (TI)	900 ms
Flip angle	9°
Voxel spacing	$1 \times 1 \times 1$ mm^3

Table 3: Acquisition parameters — Hvidovre Hospital dataset.

5.2 Preprocessing

The suggested preprocessing pipeline described in this section replicates, step-for-step, the procedure reported by Guha et al. [4] and was applied identically across al experiments condtion and both datasets. The pipeline comprises five sequatial operations applied to each 2D coronal T1-weighted MRIslices.

(1) Grayscale conversion. Each RGB JPEG image was converted to a single channel grayscale representation. This step reduces computational complexity in following operations and is a prerequisite for the grayscale specific filters that follow[4]. (2) Min-max normalization. Pixel intensities within each image were linearly rescaled to the interval $[0, 1]$ on a per iamge basis. This standardises the range across scans acquired under heterogeneous imaging condition, ensuring that diffences in scanner gain or acquisition protocol do not introduce systematix variation into the feature representation presented to the classifier [4]. (3) Contrast Limited Adaptive Histogram Equalisation (CLAHE). Local contrast was enhanced using CLAHE [10], with a clip limit of 2.0

and a grid szie of 8 × 8 pizels. Unlike global histogram equalisation, CLAHE subdivides the image into non-overlapping contextual regions and applies histogram equalisation independently within each grid, enhancing local contrast variation that would otherwise be supperesed by global equalisation methods. The clip limit bounds the amplification applied to any histogram bin, preventing over amplification of noise characteristic of unconstrained adaptive methods[10]. For MRI of cortical architectures this localized enhancement should improves the visibility of fine structural details such as gyral folding patterns and gray-white matter boundaries [4] [5]. (4) Bilateral filtering. Noise reduction was achived with bilateral filter [8], with a kernel size of 9 × 9 pixels and $\sigma_{color} = \sigma_{space} = 75$. The bilateral filter replaces each pixels intensity with a weighted average of its spatial neighbourhood, where contributions are controlled both by a Gaussian spatial kernal and a Gaussian range kernel [8]. This dual-weighting mechanism suprresses high-frequency noise in homogeneous regions while preserving sharp trasitions at anatomical boundaries – a property important for retaining cortical surfaces structure[2]. Following Guha et al [4], who evalauted the bilateral filter agaist Non-Local Means, Anisotropic Diffusion, and Wavelet Denosing, the bilateral filter was selcted as the most effective method for preserving the coritcal boundaries and structure. (5) Canny edge detection. The Canny edge detector computes a edge map [3], with a low threshold 50, a high threshold of 200, and a Sobel aperture size of 3. The canny algoithm applies a multi-stage pipeline comprosong Gaussian smoothinh, gradient magnitude estimation, non-maximum suppression and hysteresis-bases edge tracking, yielding thin, well localized edge contours with reliable boise rejection [3]. For PMG detection, accurate delineation of subtle cortical irregularities, including irregular gyri and abnormally shallow sulci, depends on precise edge localisation. The Canny detector was selcted over the Sobel operation, Scharr operator , and Laplacian of Gaussian filter after avlaution, all of which preduced less reliable results on cortical MRI, acordding to guha et al. [4]. THe resulting binary edge map was blended with bilaterally filtered image a $\alpha = 0.20$, subtly reinforcing structural boundaries without discarding the underlying intensity information.

5.3 Models and Training

Two pretrained deep convolutional neural networks architectures were evaluated, following Guha et al. (2025): ResNet-101 (He et al., 2016)ad real reff and DenseNet-201 (Huang et al., 2017), both initialized with ImageNet weights. The final fully connected layer of each model was replaced by a two latyer classification head: a linear projection to 256 unit, ReLU activation, dropout (p= 0.5), and a final linear layer to a single logits, e.g. PMG vs control, following the head design of Guha et al.(2025). The backbone was frozen and only the classification head was trained, following Guha et al. (2025). Input images were resized to 224 ×224 pixels and replicated across three channels to match the ImageNet input format. Standard data augmentation was applied during training: random resized crop, 80-100% of the image area, resized to 224 ×224 pixels, and random horizontal flip.

Models were trained using the Adam optimizer with a learning rate of 5×10^{-4}, weight decay of 1×10^{-3}, and a batch size of 32, for 20 epochs, with no early stopping. Binary cross-entropy with logits (BCEWithLogitsLoss) was used as the loss function across all conditions. Performance was evaluated using accuracy, precision, recall, F1 score and Cohens k.

5.4 Ablation study

To probe whether trained models rely on spatially localised image features or on global low-level statistics, a black-box acclusion ablation was applied to all retained checkpoints. For each test imag, a random square patch with a side legnth equal to 20% of the shorter image dimension was zeroed out prior to inference. The occluded test set was constructed once per split and then passed through every available checkpoint. Classification metrcis were computed on the occluded images and compared to the correspondsing unoccluded test perfermance. A substantial drop in performance under occlussion is consistnt with the model attending to specifc spatial regions; a negligible drop suggest reliance on global texture or low-level statistics distributed across the image, which would further support the shortcur.learning hypothesis.

6 Experimental setup

To evaluate the methodological validity of Guha et al. (2025) and isolate the contribution of each identified error, experiments were organised into seven conditions spanning two axes: label definition (paper vs. corrected) and downsampled strategy (none, pre-split, post-split). Both ResNet-101 adn DenseNet-201 were evaluated. Five-fold cross validation at the patient level was applied to two condition: preprocessed data with paper labels and preprocessed data with corrected labels, condition 3 and 4, respectively.

Following Guha et al. (2025), the replication condition treated all 4517 slices attributed to PMG patients as the positive class, regardless of their slice-level annotations (label = 1, 2, or 3). The 10539 control slices were globally downsampled to 4517 before splitting, matching the class counts reported in their paper. Data was then split into training (60%), validation (20%), and test (20%) sets. The corrected label definition restricts the positive class to slices with label = 1 (n = 2256). Label = 2 (n = 1386) are reclassified as negative examples and pooled with the original healthy controls, yielding 10539 + 1386 = 11925 negative slices. Label = 3 slices are excluded from corrected conditions. All splits in the conditions weree performed at the patient level: the 92 unique subject were paritioned such that all slices from a given patient appear exclusively in one split. A fixed random seed (seed = 42) ensured reproducibility.

The experimental setup developed throughout the project. With further investigation into

Guha et al. new experimental condition was developed. Overall seven condition were tested, structued to allow pairwise comparison of each error in isolation:

- Condition 1 – Raw data, paper labels: Models trained on raw data, unpreprocessed, with the paper label definition (4517 PMG, 10539 controls). No Downsampling applied.
- Condition 2 – Raw data, corrected labels: Models trained on raw data with the corrected label definition (2256 PMG, 10925 controls), patient-level split. No downsampling applied.
- Condition 3 – Preprocessed, paper labels: Models trained on preprocessed data with the paper label definition. No downsampling applied. Comparing with condition 1 isolates the effect of the preprocessing pipeline under the paper label definition.
- Condition 4 – Preprocessed, corrected labels: Models trained on preprocessed images with the corrected label definition, patient-level split. No downsampling applied. Comparing with condition 2 isolateds the effect of the preprovessing under the corrected label definition.
- Condition 5 – Preprocessed, paper labels, pre-split downsampling: Models trained on preprocessed data with the paper label definition and the healthy control class globally downsampled to 4517 slices before splitting, matching Guha et al.'s setup. Comparing with condition 3 isolates the effects of the pre-split downsampling under the paper label definition.
- Condition 6 – Preprocessed, corrected labels, pre-split downsampling: Models trained on preprocessed data with corrected label definition but with pre-split downsampling of the healthy control. Comparing with condition 5 isolates the effect of label correction when downsampling strategy is held fixed.
- Condition 7 – Preprocessed, corrected labels, post-split downsampling: Models trained on preprocessed data with the corrected label definition and downsampling applied only within the training parition after splitting. Validation and test sets retain the natural 1:5.3 class ratio (2256:11925). This is the fully corrected condtion. Maybe i should train raw, corrected labels and post-split downsampling as the fully corrected? ask llucia/mmelanie. Comparing with condition 6 it isolates the effect of downsampling timinng under the corrected label defnition.

Then 5-fold cross validation at the patient level was used to obtain stable performance estimates for condition 3 and 4, to compare with Guha et al. results. Patients were divided by class and split into 5 folds, ensuring approximately equal calss representation per fold. Withing each fold, 15 % of the non-test patients were as a validation set; the models checkpoint with lowest validation loss was saved and evaluated on the test fold.

7 Results

7.1 PPMR dataset

A second confounding factor is the systematic difference in image resolution between classes. As shown in Figure 3, PMG-patient images are exported from the PACS system at approximately 1508×1727 pixels, whereas healthy control images range from 260×320 pixels to 512×512 pixels. When all images are resized to the 224×224 pixel model input, PMG images are compressed by a factor of approximately $6\times$, while HC images are resized by only 1.1–$2.3\times$. Heavy compression averages many original pixels into each output pixel, producing smooth, low-variance texture. Near-native resizing preserves the coarser, higher-variance texture of the original acquisition. A classifier can therefore distinguish the two classes simply by learning that smooth texture implies PMG and coarser texture implies HC – a signal that is present in every patch of every PMG-patient image, regardless of whether cortical pathology is visible in that region.

3.

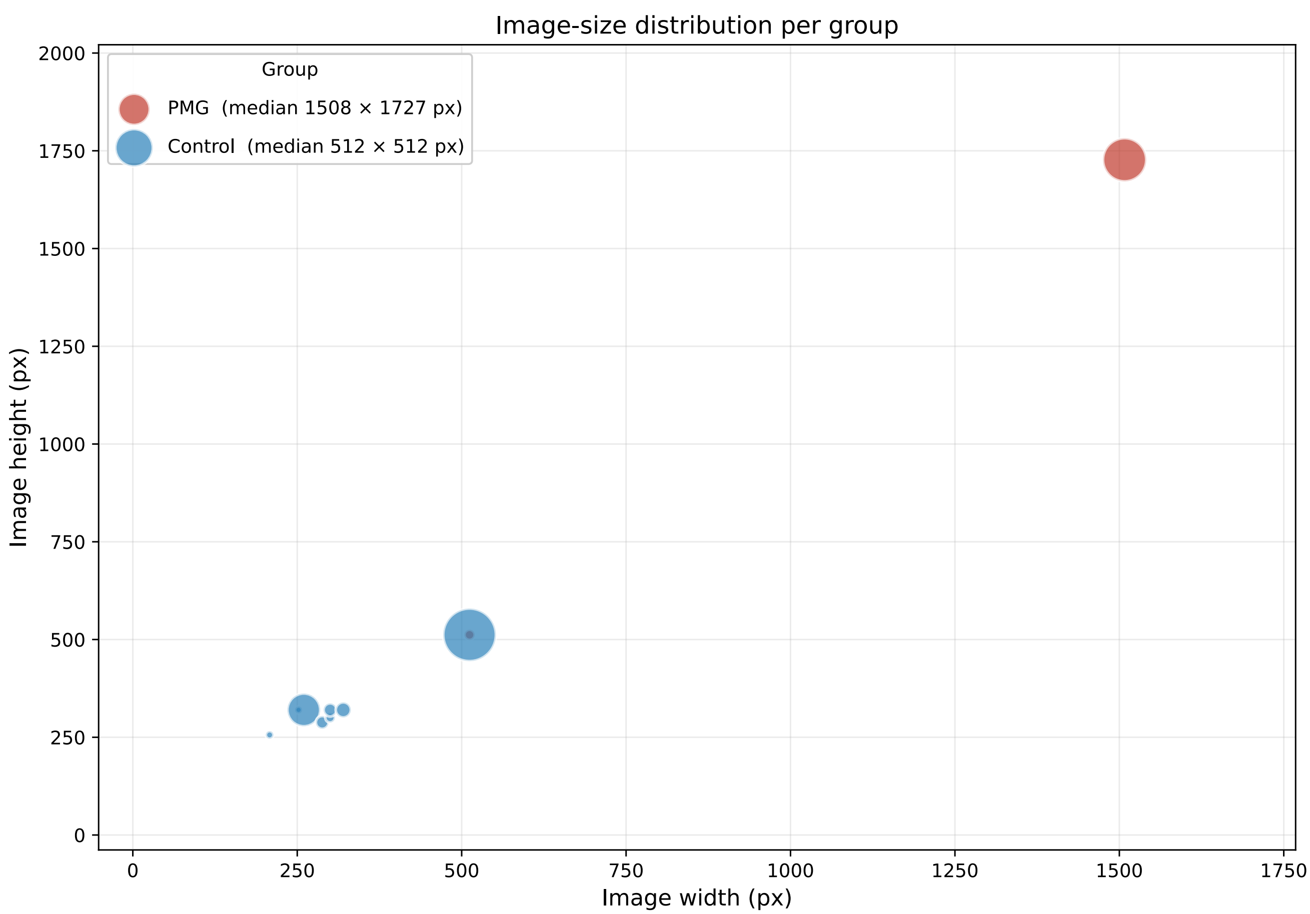

Figure 3: Caption

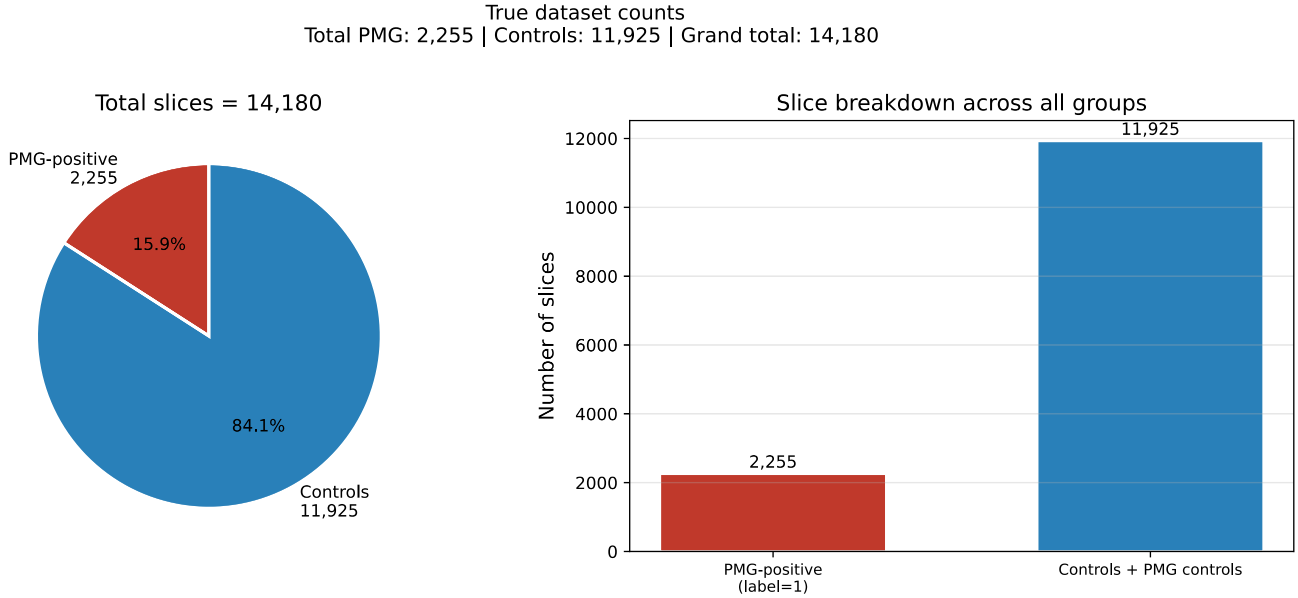
True dataset counts
Total PMG: 2,255 | Controls: 11,925 | Grand total: 14,180
Total slices = 14,180
PMG-positive
2,255
15.9%
84.1%
Controls
11,925
Slice breakdown across all groups
Number of slices
12000
10000
8000
6000
4000
2000
0
11,925
2,255
PMG-positive
(label=1)
Controls + PMG controls

Figure 4:

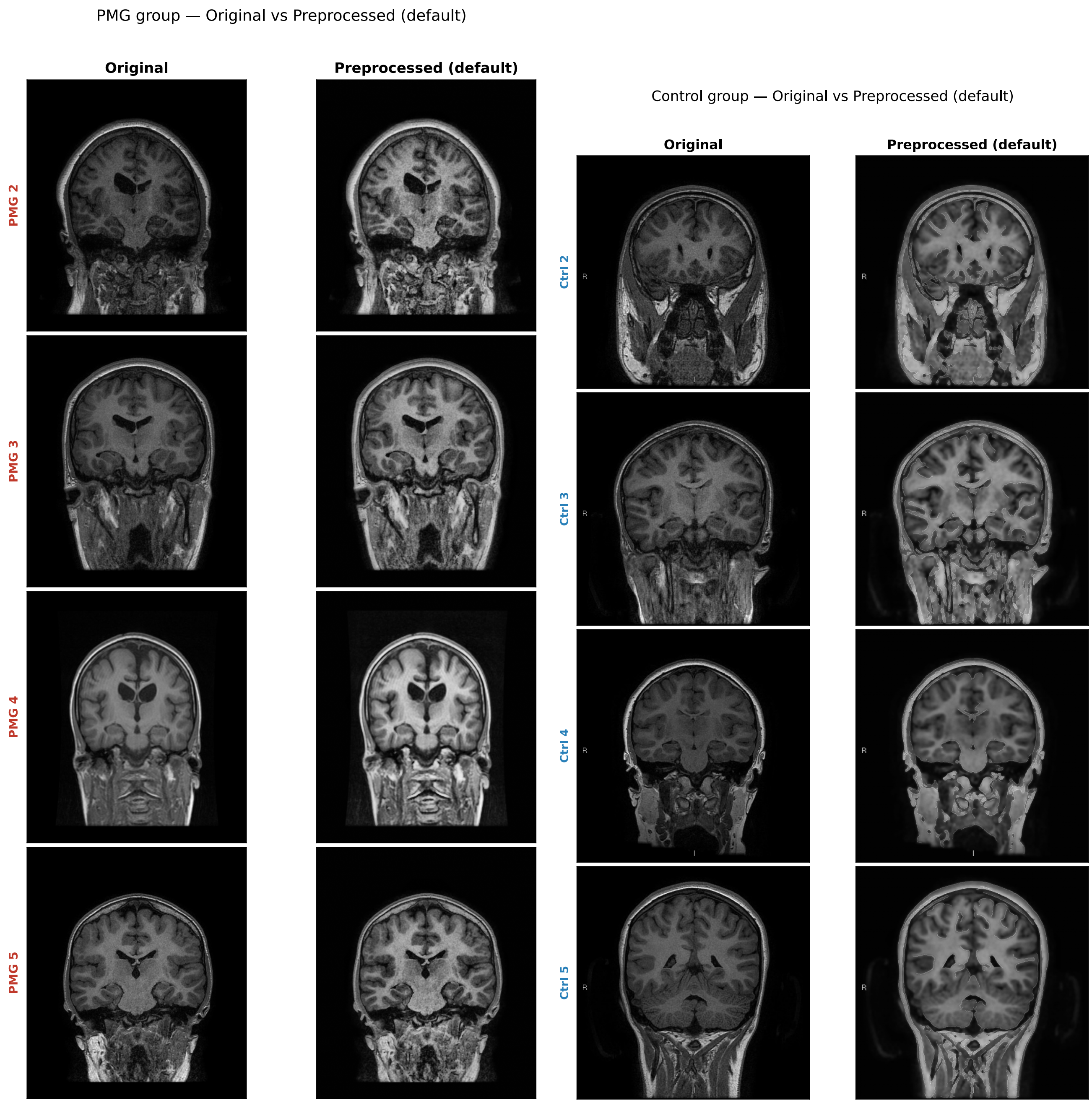

Figure 5: Caption

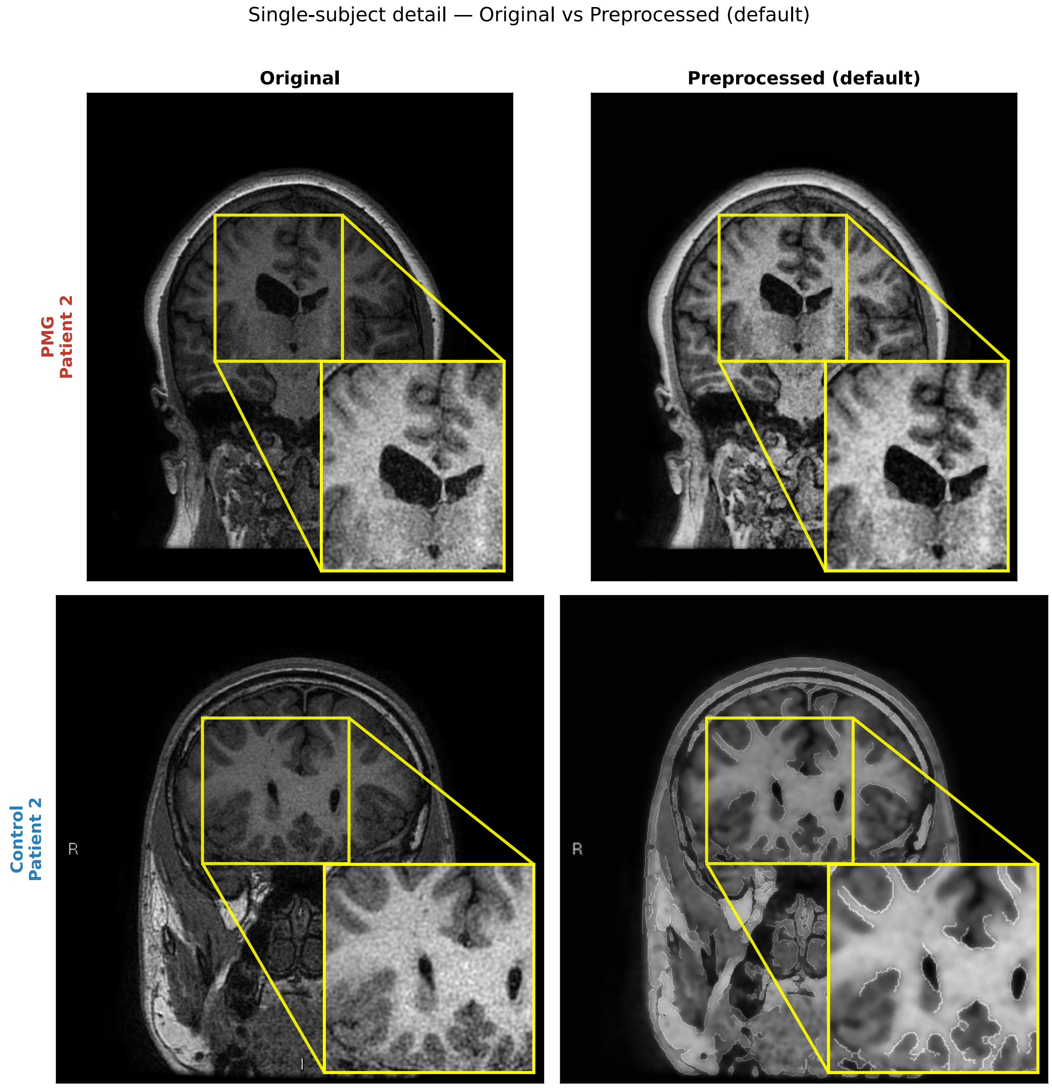

Figure 6: Caption

8 Discussion

9 Conclusion

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